Final Progress Report

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Project : Automated Food ingredient safety classification using ocr and machine learning.

Progress summary:

I have made substantial progress on all major components of the project. The system now functions as a complete end-to-end pipeline that takes a food label image as input and produces a safety classification for each detected ingredient. an end-to-end pipeline that classifies food ingredients as **Safe**, **Moderate**, or **Harmful** using a combination of Optical Character Recognition (OCR), custom ingredient parsing, fuzzy matching against a curated database, and supervised machine learning classifiers.

Pipeline Architecture

The system follows a four-stage pipeline:

1. **OCR Stage** — EasyOCR extracts raw text from a food label image.
2. **Parsing Stage** — A custom regex-based parser isolates individual ingredients, handling sub-ingredients in parentheses, OCR artifacts, and common label formatting variations.
3. **Matching & Classification Stage** — Each parsed ingredient is first matched against a known database via fuzzy string matching. If no match is found, it falls back to a trained ML classifier.
4. **Visualization & Summary Stage** — Results are presented as a per-ingredient safety table and visual charts (pie chart + horizontal bar chart)

Database

The dataset (dataset.xlsx) was compiled from three sources:

- **CSPI (Center for Science in the Public Interest)** — Chemical Cuisine database
- **FDA (U.S. Food & Drug Administration)** — GRAS (Generally Recognized as Safe) listings
- **Open Food Facts** — Community-sourced ingredient data

The dataset exhibits significant **class imbalance**, with "Safe" being the dominant category. This was addressed through stratified train/test splitting and stratified cross-validation. The ingredient safety database remains at 710 entries sourced from CSPI Chemical Cuisine, FDA databases (GRAS list and food additive status listings), and manual collection from Open Food Facts. I implemented a text preprocessing module that cleans and normalizes ingredient names, descriptions, and synonyms, then combines them into a single feature vector per ingredient. The

TF-IDF vectorizer converts these combined text features into numerical representations using unigrams and bigrams

Machine Learning Models:

I trained and evaluated three classifiers Naive Bayes, Support Vector Machine (SVM), and Decision Tree using stratified 5-fold cross-validation on an 80/20 train-test split. The evaluation includes accuracy scores, full classification reports (precision, recall, F1-score per class), and confusion matrices for each model. A comparison summary identifies the best-performing model, which is then used later . The **Decision Tree** was selected as the best model for the final pipeline.

Naive Bayes:

High precision but low recall for minority classes — the model is conservative, rarely predicting "Harmful" or "Moderate" but correct when it does. The class imbalance biases it toward predicting "Safe."

Support Vector Machine (SVM):

Substantial improvement over Naive Bayes. Better recall on minority classes, indicating the linear decision boundary captures more nuanced patterns in the TF-IDF space.

Decision Tree:

Best overall performance. Notably, it achieves 0.97 recall on "Moderate" — critical for consumer safety since moderate-risk ingredients should not be missed. The macro F1 of 0.91 indicates strong balanced performance across all classes.

I integrated EasyOCR for text extraction from food label images. To improve OCR accuracy on real-world labels with varying font sizes, contrast, and curvature, One of the most challenging aspects of this project has been parsing the raw OCR output into individual ingredient names. I developed an Ingredient Parser class that handles a wide range of real-world edge cases, including multi-section labels , nested parentheses and brackets etc.

Parsing

The parser follows a multi-step process:

1. **Keyword detection:** Searches for ingredients: or contains: markers to isolate the ingredient section from surrounding nutritional/label text.
2. **Noise removal:** Strips non-ingredient phrases (e.g., "nutrition facts," "serving size," "manufactured by").
3. **Parenthetical sub-ingredient extraction:** A custom _extract_sub_ingredients() function uses depth-tracking to find parenthetical groups, then applies a heuristic (_is_sub_ingredient_list()) to distinguish sub-ingredient lists from descriptors.

4. **Splitting:** Ingredients are split on commas, semicolons, periods, and the word "and."
5. **Cleaning:** Each token is lowercased, stripped of non-alphabetic characters, and deduplicated.

Sub-Ingredient Heuristic

The _is_sub_ingredient_list() function classifies parenthetical content as either:

- **Sub-ingredient list** (e.g., "corn meal, ferrous sulfate, niacin") - if it contains commas or is 1–4 words without descriptor prefixes.
- **Descriptor** (e.g., "made from concentrate", "e260") - if it starts with known descriptor words or matches an E-number pattern.

The classification module uses a three-strategy approach: first attempting an exact or fuzzy database match, then checking against the synonym lists, and finally falling back to the trained ML model for any unrecognized ingredients.

Fuzzy matching:

Each parsed ingredient is matched against the known database using a three-tier strategy:

1. **Exact match:** Direct string equality - confidence 1.0
2. **Substring match:** If either string contains the other -confidence 0.9
3. **Fuzzy match:** SequenceMatcher ratio - accepted if ≥ 0.75 threshold

I tested the pipeline on multiple label types, including a basic beverage label, a protein bar, a multi-section ramen package (maggi) etc. The system generates pie charts showing the safety category breakdown and horizontal bar charts showing each ingredient color-coded by classification.

Visualization

The system generates two visualizations per analysis:

1. **Pie chart** : Shows the proportion of Safe/Moderate/Harmful ingredients in the product.
2. **Horizontal bar chart** :Lists each ingredient color-coded by safety category (green = Safe, orange = Moderate, red = Harmful).